ADVANCED BUSINESS ANALYTICS

AI-Powered Telecom Customer Retention

A staff-approved workflow that turns churn alerts into coordinated customer care.

Presented by:

Group 3

OPENING

PDMT Framework

PROBLEM

- **Rising churn**
  - National Internet churn +20% YTD
  - Binh Duong churn increased: 1,300 (Aug) → 2,500 (Sep)
- **Alert-to-action gap:** Staff see risk scores but are unsure how to approach customers
- **Late contact and unresolved issues**

DATA

- **Customer Data:** Profile, plan, billing, service, usage and support history
- **Conversation Data:** Messages, replies, approvals, follow-ups and case outcomes
- **ML Model & SHAP Result:** Churn scores and the risk drivers behind each score

METHOD

- **ML churn prediction:** Compare LightGBM and CatBoost → Select tuned LightGBM
- **SHAP explanation:** Group risk drivers into six business categories
- **AI Agents**
  - explain the case
  - draft messages, write call scripts
  - draft support tickets from chats

TOOLS

- **Python + ML Libraries**
- **Supabase / PostgreSQL**
- **React / Vite**
- **n8n**
- **LLM + Demo Knowledge Base**

From prediction to approved customer action.

PROBLEM

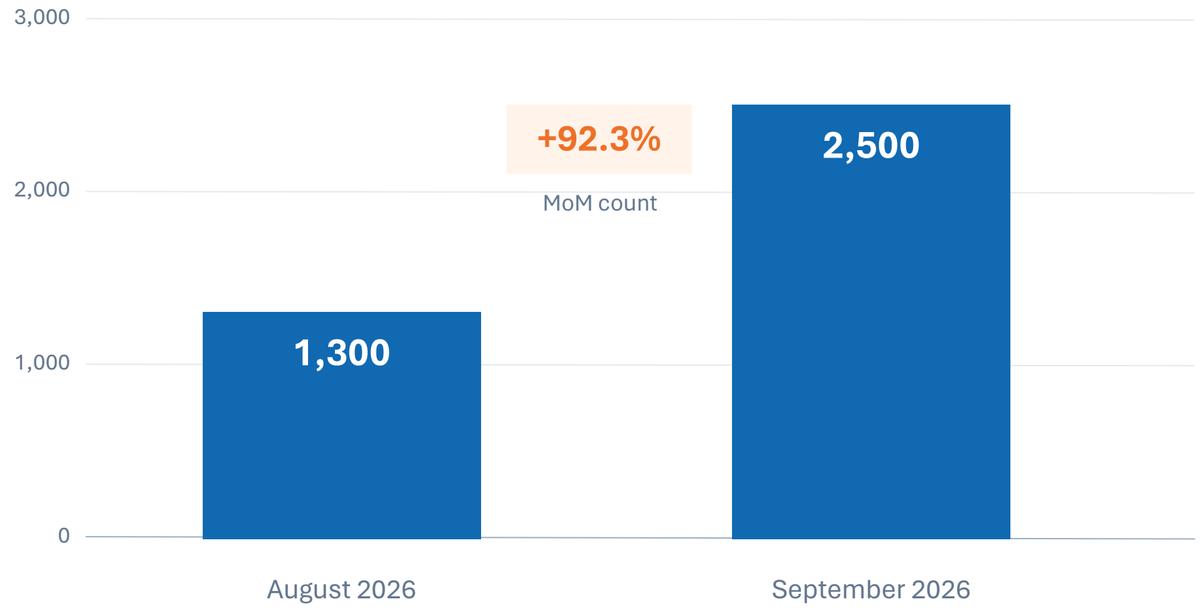
Business Context and Retention Opportunity

FPT Telecom

Internet services

Binh Duong - September Churn Spotlight

Reported monthly customer churn



Reported as the largest increase in churn rate;

+20%

National Internet churn count

YTD 2026 vs. comparable 2025 period

1.19%

September churn rate · Binh Duong

2,500 / approximately 210,000 customers

Estimated subscription revenue exposure · Binh Duong only

1 MONTH	6 MONTHS	12 MONTHS
625M	3.75B	7.5B

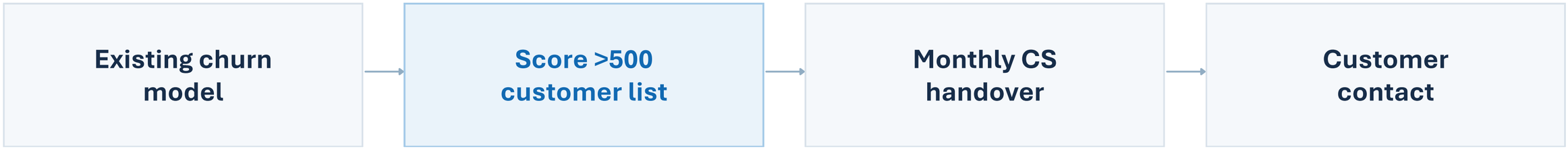
VND · same 2,500-customer cohort; VND 250,000/month (assumed national average ARPU). 6/12-month values assume no return and unchanged fee.

Higher churn increases the need for timely, accountable customer care.

PROBLEM

Current Process and Operational Gaps

Reported current process



Three reported operational gaps

Reported gap	Operational difficulty	Decision needed
Unclear approach	Risk signal without enough context	Why might this customer churn?
Late contact	Some customers churn before contact	Who needs attention first?
Unresolved issue	Contact may not address the concern	What action is needed?

Risk alerts alone do not ensure timely, appropriate or effective care.

PROBLEM

Stakeholders and Decisions

Four groups support the customer-care workflow

Stakeholder	Key decision	Information needed	Expected value
CS Agent	Choose approach and action	Customer facts · SHAP · history	Relevant customer care
CS Supervisor	Assign and monitor cases	Queue · ownership · progress	Clear accountability
Data Analytics	Evaluate prediction utility	Model version · score · outcome	Traceable feedback
Technical Support	Investigate service issues	Symptoms · support requests	Appropriate handoff

DIRECT USERS **CS Agent & CS Supervisor**

COORDINATION **Data Analytics & Technical Support**

Staff decide and act; supporting functions provide evidence and escalation.

PROBLEM

Project Objectives and Scope

Three operating objectives

01 Prioritize and understand

Review churn alerts with customer facts and model explanations.

02 Prepare and coordinate

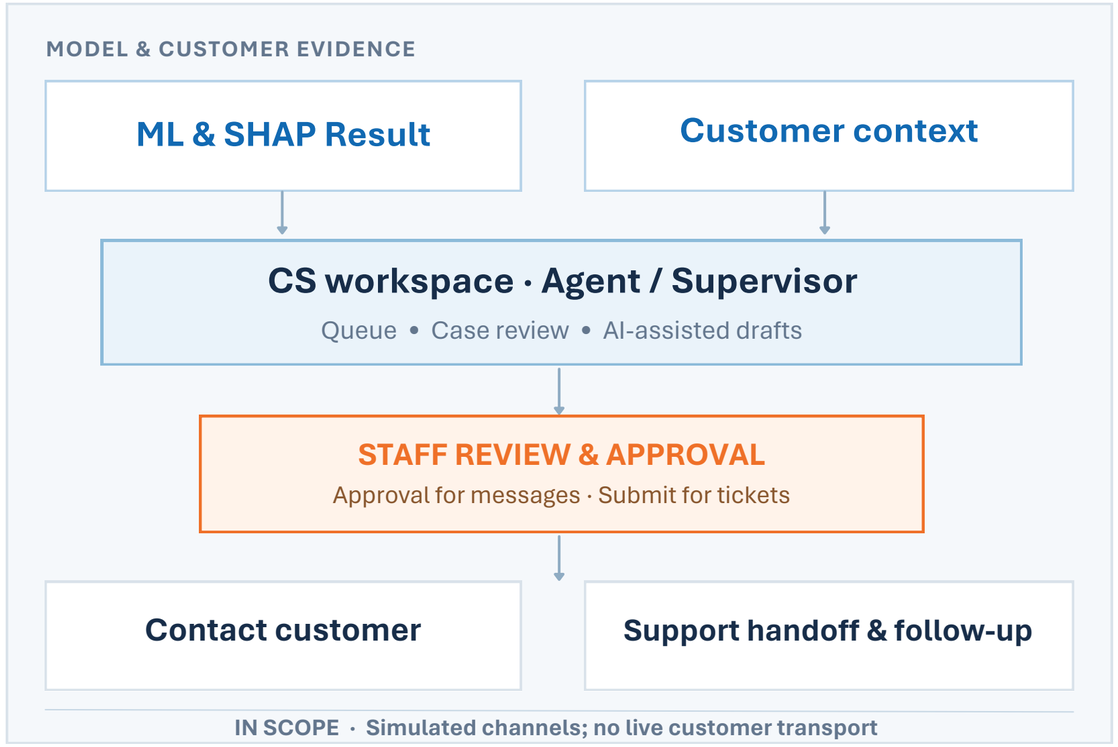
Use AI drafts, call scripts and support tickets to prepare the next action.

03 Preserve accountability

Track ownership, human approval, follow-up and period outcomes.

AI assists staff; it does not act autonomously.

Prototype boundary



DATA

Data Requirements and Sources

What comes from Kaggle — and what the prototype synthesizes

Data source	Available information	Role in the prototype
01 Kaggle Dataset	Customer demographics, plan / contract, service flags, billing attributes and churn fields	Public fictional source; 7,043 customers
02 Synthetic ML data	Generated service, billing and support histories; as-of snapshots; 127 model features	ML training, churn scoring and SHAP explanations
03 Prototype / AI records	AI drafts, mock conversations, support tickets, approvals and follow-up	Simulated customer-care workflow

ORIGINAL KAGGLE

7,043

customers · 38 columns

INTERNET COHORT

5,517

customer seeds

SYNTHETIC DATA

113,937

as-of snapshots · 127 features

Different grains: source customers → Internet cohort → repeated snapshots → one scoring batch.

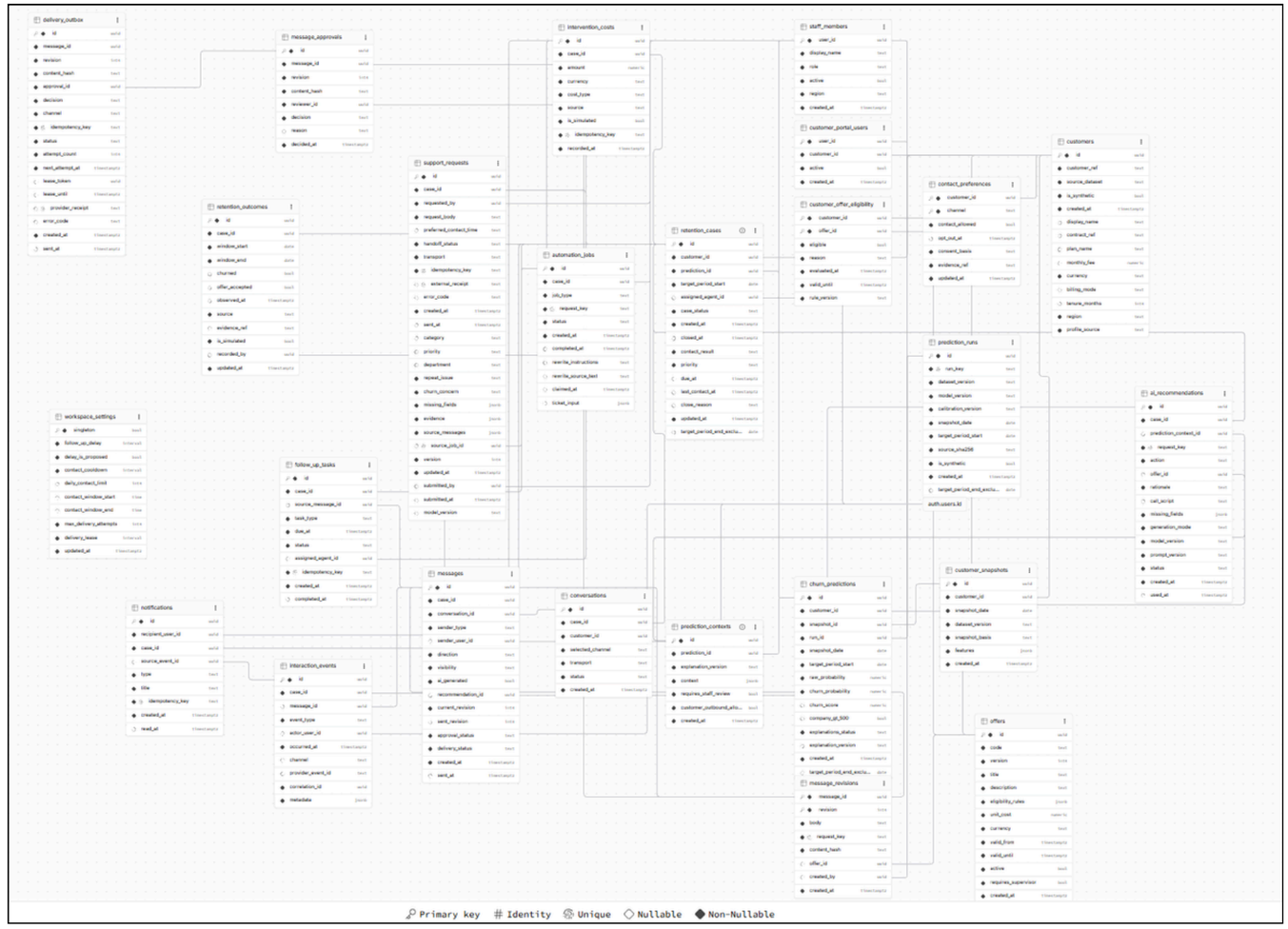
Data Model — Supabase

Database structure and entity relationships

SCHEMA SUMMARY

28 tables

Core domains:
Customer & ML · Cases & Messaging · AI · Support & Outcomes



DATA

Example Customer Snapshot Record

One customer · One as-of date · Selected fields from 127 input features

CUSTOMER 0196-VULGZ	CONTRACT DEMO-0196-VULGZ	SNAPSHOT 10 OCT 2026

01 CUSTOMER & SUBSCRIPTION

Contract	DEMO-0196-VULGZ
Internet plan	Demo Internet 500 Mbps
Tenure	15 months
Monthly charge	VND 300,000

02 USAGE & SERVICE

Usage (30 days)	284.00 GB
Disconnections (30d)	1
Downtime (30d)	15.26 minutes
Support contacts (30d)	0

03 BILLING & PAYMENT

Billing mode	Postpaid
Fee change (30d)	0%
Current overdue amount	VND 600,000
Longest overdue delay	30 days

04 CHURN PREDICTION

Calibrated probability	**60.67%**
Risk score	**606.74 / 1,000**
Selected model	LightGBM tuned
Observed churn	**Unknown**

PREDICTION WINDOW	11 October–10 November 2026	Synthetic demo record · Not a real FPT customer

METHOD

Literature Review: Approaches to Customer Retention

Four research streams answer four different decisions

01 CHURN PREDICTION & EXPLAINABILITY

Who may leave?

Customer data → Risk score → SHAP factors

WHAT RESEARCH SHOWS

Models estimate who may leave; SHAP shows which details shaped that estimate

LIMITATION

Explaining risk does not prove which action prevents churn.

Poudel et al. (2024) · Asif et al. (2025) · Lundberg et al. (2020)

02 PROFIT-DRIVEN RETENTION

Is contact worth the cost?

Customer value – Campaign cost

↓ Expected net gain ↓

WHAT RESEARCH SHOWS

Profit models select customers by expected net return, not risk alone.

LIMITATION

The calculation needs real customer value, contact costs and response rates.

Liu et al. (2024) · Jiang et al. (2024) · Gómez-Varga et al. (2025)

03 TARGETING BY INTERVENTION EFFECT

Who changes if contacted ?

CONTACTED GROUP Retention – CONTROL GROUP Retention

Difference = estimated uplift

WHAT RESEARCH SHOWS

Uplift compares outcomes with and without an intervention.

LIMITATION

Needs contacted and comparison groups; the result may be uncertain.

Ascarza (2018) · Bokelmann & Lessmann (2024) · Verhelst et al. (2025)

04 AI ASSISTANCE FOR SERVICE STAFF

How should staff act?

AI suggestion → Human review → Customer contact

WHAT RESEARCH SHOWS

A field study found faster service with AI; benefits differ by worker and task.

LIMITATION

Check service quality and staff workload in this setting.

Gnewuch et al. (2024) · Vaccaro et al. (2024) · Brynjolfsson et al. (2025)

KEY TAKEAWAY

A risk score helps find customers to review. The team must also consider value, likely response and staff judgment.

METHOD

Literature-Informed Design Choices

LITERATURE LIMITATIONS THAT SHAPED THE DESIGN

01 **A model can identify risk factors, but these factors do not prove what caused the customer to leave**

02 **A risk score identifies a problem, but it does not assign responsibility or define the next action.**

03 **AI suggestions may be incomplete or unsuitable for the customer's situation due to missing context**

04 **Contact alone does not does not prove that the customer was retained.**

HOW THE PROJECT HANDLES EACH LIMITATION

01 / EXPLAIN RISK

Churn model + risk factors

127 inputs → 6 risk areas
Signals from the model, not proven causes

Staff can review the evidence behind the alert without treating the explanation as proof of causation.

02 / CREATE A CASE

Customer record + care queue

Risk alert → assigned case
Customer details and next step

Every alert enters a trackable workflow instead of remaining as an isolated prediction.

03 / ASSIST STAFF

AI suggestions + Human Decisions

Explanation · Message · Call guide
Staff checks each suggestion before contact

AI reduces preparation work while the final customer-facing decision remains with staff.

04 / RECORD ACTION

Staff approval + Contact

Review → Send → Reply
Result and cost recorded in the system

The case keeps messages, replies and follow-up history.

DESIGN TAKEAWAY

The project turns a risk alert into a clear care process: explain the risk, assign the case, review AI suggestions, and record what happened.

METHOD

Analytical and Decision Workflow

From risk score to customer care: Customer cases moves through six steps. AI supports preparation, while staff control the customer-facing action.

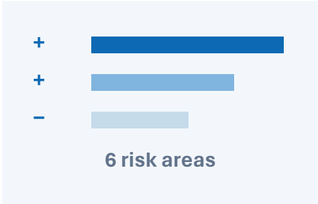
01
Identify



The churn model identifies a high-risk customer.

OUTPUT
Risk score + time period

02
Understand



The system presents the main risk areas together with relevant customer information

OUTPUT
Risk factors + customer details

03
Prepare



AI uses available information to prepare message and call guides to care customers.

OUTPUT
Suggested message + contact checks

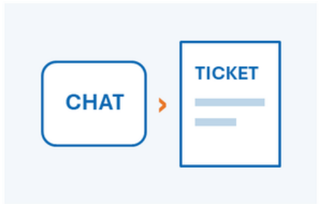
04
Approve



Staff check evidence, edit the draft and decide whether to contact the customer.

OUTPUT
Approved action or no-contact decision

05
Coordinate



Record the exchange and any technical handoff. AI drafts a ticket for Technical department if needed.

OUTPUT
Interaction history + Ticket draft

06
Evaluate



Track follow-up actions and check the customer's service outcome after the full assessment period

OUTPUT
Retention result + cost

HUMAN CONTROL

AI may prepare recommendations, but staff must review and approve every customer-facing action before it is sent.

AFTER CONTACT / MEASURE RESULTS

A reply or closed case alone does not show that the customer stayed. Retention is evaluated separately using the customer's later service outcome

METHOD

Three AI Agents Assist Staff at Each Step

Each agent drafts from approved facts. Staff review and decide every customer-facing action.

1 AI Explanation Agent

Step 2 · Understand

INPUT

SHAP risk groups, plan, billing, usage and support facts

OUTPUT

Plain-English explanation of the top risk signals, with a suggested action

Explains the score; never changes it or proves the cause

2 AI Message & Call Agent

Step 3 · Prepare

INPUT

Case status, chat history, staff request and approved FAQ

OUTPUT

Chat draft, call script and next-step guidance for staff

Staff check facts, tone and consent; delivery is simulated

3 AI Ticket Agent

Step 5 · Coordinate

INPUT

Selected chat with message IDs and timestamps

OUTPUT

Editable ticket draft with chat evidence and missing details

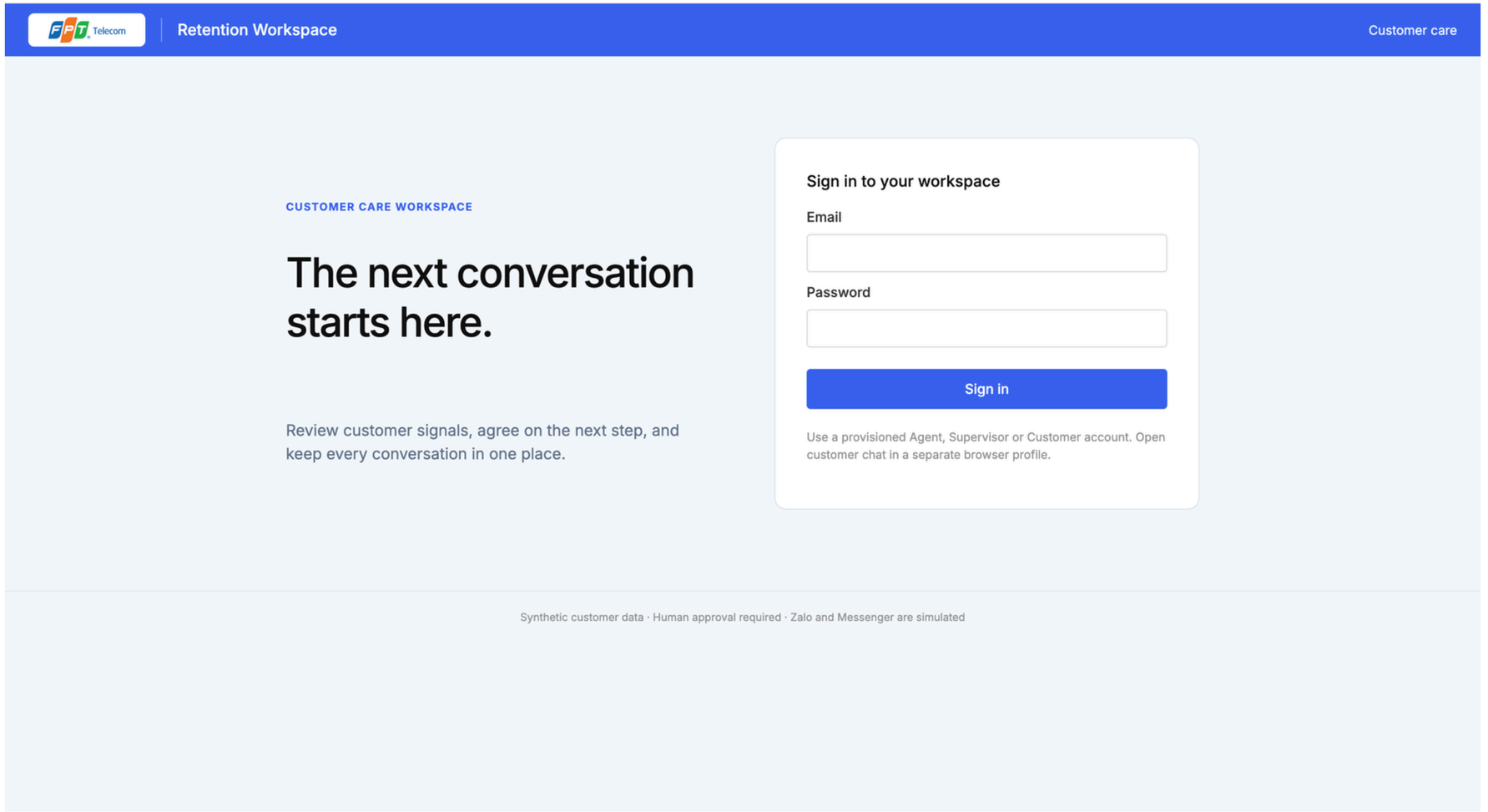
Blocked without a chat; staff decide when to submit

AI prepares. Staff approve. No AI action reaches a customer without human review.

METHOD

Prototype Demonstration

https://ai-powered-telecom-customer-retention.vercel.app/



ACCOUNT & PASSWORD

Supervisor

supervisor@retention.example

Agent

agent@retention.example

Customer

customer@retention.example

Password

123456

Video demo

METHOD

Data and Evaluation Design

Component	Description
Input	**127 features per snapshot**, using only information available at the cutoff. IDs and target labels are excluded.
Snapshots	**3 each months:** Previous month-end (28/29/30/31), day 10 and day 20; predictions issue on days 1, 11 and 21 to refresh customer risk.
Target	Churn within the **next ~30 days.**
Temporal split	**January**-start windows for **training**; **July**-start windows for **validation**; **September**-start windows for **testing**.
Leakage control	**Train only on outcomes fully observed before validation begins**. Reserve separate customers for probability calibration and audit testing.



Example: Snapshot 10 October → prediction end 11 October → target window [11 October, 11 November).

Why time-series splitting? Random row splitting can mix future and past information and separate overlapping customer records across sets, producing optimistic evaluation results.

METHOD

Model Training and Selection

Why LightGBM and CatBoost?

Both are gradient-boosted tree classifiers that capture nonlinear relationships and feature interactions in tabular customer data. Both accept categorical inputs and use the same 127 features, allowing a consistent comparison.

Optuna TPE - finding suitable hyperparameters

Optuna tests 30 parameter combinations per model family, including learning rate, tree depth and regularisation. It uses earlier trial results to propose promising combinations. Each trial is evaluated across three chronological folds to maximise mean ROC AUC.

Platt Calibration - making probabilities more reliable

A model can rank customers well but overestimate or underestimate their churn probabilities. Calibration adjusts those probabilities using a separate customer group. We adopt the adjustment only when both Brier score and log loss improve in out-of-fold evaluation.

Candidate	Validation AUC	Test AUC	Test AP
LightGBM baseline	0.9469	0.9473	0.7166
LightGBM tuned	0.9483	0.9467	0.7138
CatBoost baseline	0.9365	0.9374	0.7183
CatBoost tuned	0.9356	0.9348	0.7253

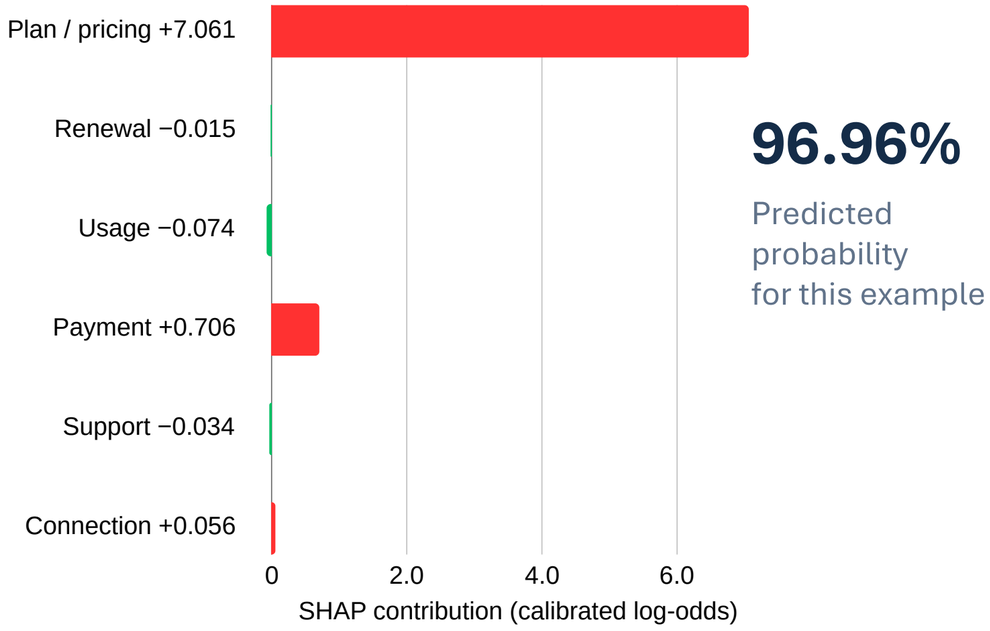
LightGBM tuned was chosen as it has the **highest validation ROC AUC** under the predefined selection rule.
Selected configuration: 114 boosting rounds, maximum depth 7 and 12 leaves.

METHOD

Risk Explanations: SHAP + AI AGENT

SHAP explains how customer information contributes to the model's predicted churn risk.

We aggregate the signed SHAP values of 127 features into six business groups in the example below.



96.96%

Predicted probability for this example

Plan and pricing contributes most to the predicted risk. Positive values increase predicted risk; negative values reduce it relative to the model baseline.

AI interpretation (View n8n workflow in Appendix D)

Component	Description
Purpose	Explain the main predicted risk drivers using concrete customer facts.
Input	SHAP contributions, customer profile, plan and billing details, service usage, and interaction history.
Output	A clear explanation connecting the strongest model signals to observed customer facts, with a suggested action for staff review.
Example	Plan and pricing is the strongest risk driver. The customer's monthly fee increased by 96.17% over 60 days, reaching VND 563,000, and they requested three plan changes in the past 30 days. These facts suggest possible concerns about pricing or plan suitability. Suggested action: clarify the fee increase and discuss eligible alternative plans.

The AI agent explains the existing prediction; it does not change the score or confirm the causes of churn.

SHAP explains the prediction. Customer facts support interpretation; the actual churn reason requires customer confirmation.

METHOD

AI Support Chat & Call

Customer facts and conversation history guide staff-reviewed communication. (View n8n workflow in Appendix D)

INPUT

Customer context

Plan, monthly fee, billing mode and tenure

Care context

Driver groups, dated facts and current case status

Conversation & staff request

Sent messages and replies, output type and rewrite request

AI PROCESSING

Check evidence

Review facts and model signals.
Flag missing or outdated data.

Understand the conversation

Address the latest reply.
Avoid repeated questions.

Consult approved guidance

Read approved FAQ references.
Keep facts and policy unchanged.

OUTPUT

Chat draft

A concise reply tailored to the customer's latest concern

Call script

Opening, discovery questions, next steps and closing

Staff guidance

Next action, rationale and information still needed

Staff review facts, tone and contact permission before sending.

For customers at risk of leaving, an impersonal reply may deepen dissatisfaction.
→ Staff conduct calls. Message delivery is simulated. AI does not apply promotional offers.

Crolic et al. (2022); Luo et al. (2019)

METHOD

AI Ticket: Chat-to-Ticket

Convert customer conversations into staff-reviewed support ticket drafts. (View n8n workflow in Appendix D)

INPUT

Customer conversation
Sent messages and customer replies from the selected chat.

Traceable source
A snapshot with message IDs and timestamps.

Staff request
Staff select Draft ticket in the Support ticket tab.

AI PROCESSING

Summarise the issue
Extract the reported problem and supporting chat quotes.

Suggest ticket fields
Category, priority, department, repeat issue and churn concern.

Identify missing details
List the information staff should confirm with the customer.

OUTPUT

Editable ticket draft
Description, suggested fields and evidence from the chat.

Clarification message
Staff can use AI Message to draft follow-up questions.

Staff-controlled submission
Staff save the reviewed version and submit it for support.

AI creates a draft. Staff decide when to send questions or submit the ticket.

Expected value: less retyping, more complete tickets and clearer routing.

METHOD

Supervisor Tools: Dashboard & Case Assignment

Team oversight and case allocation with Supervisor review.

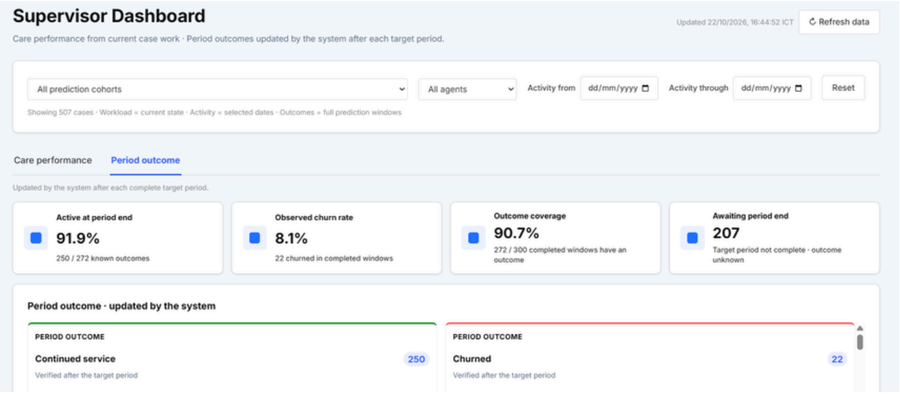
DASHBOARD

Monitor workload and care progress

View open and unassigned cases, overdue follow-ups and each Agent's workload.

Review outcomes and care issues

Compare outcomes for completed windows and open matching cases for follow-up.



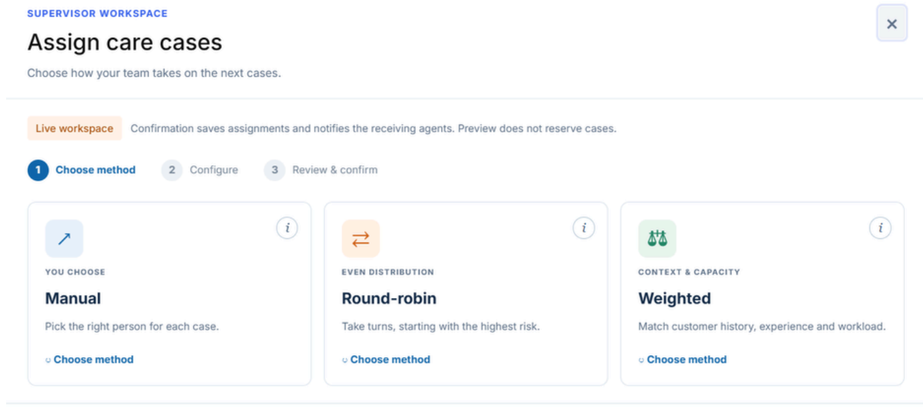
CASE ASSIGNMENT

Manual, Round-robin or Weighted

Assign directly, rotate across Agents, or weigh customer continuity, experience and workload.

Preview and confirm the allocation

Prioritise higher-risk cases within capacity.
Review proposed owners before confirming.



Supervisor controls allocation. Staff review every outbound message.

TOOLS

Technology Architecture and Tool Choices

Tool	Role in the solution	Why we chose it
Python + ML Libraries	Prepare features, train and compare models, predict churn, and calculate SHAP values.	Supports repeatable experiments and detailed model evaluation.
Supabase / PostgreSQL	Store customer data and case records, manage sign-in, and control access by role.	Keeps data consistent and limits access to authorized users.
React / Vite	Build the Agent and Supervisor interfaces and the simulated customer portal.	Supports an interactive interface for reviewing cases and managing customer care.
n8n	Connect AI requests, simulated message delivery, and case updates.	Visual workflows make each process easier to inspect and maintain.
Large Language Model + Demo Knowledge Base	Use customer facts, SHAP results, and approved guidance to draft explanations, messages, and call scripts.	Turns available information into readable drafts for staff to review.

From prediction to action: staff review AI drafts, approve messages, and track results in one system.

TOOLS

System Testing Approach

Whole-system checks across three user roles

Area	Coverage	Method
Access and privacy	Supervisor dashboard and assignment; Agent case access; Customer portal privacy.	Sign in by role; check visible data and available actions
AI assistance	Explanation, Message, Call script, Ticket and clarification drafts	Compare output with customer facts; review before use
Workflow and safeguards	Drafts, customer replies, staff approval, empty-chat protection and failed-care restrictions.	Browser scenarios plus automated UI, workflow and database checks

Customer cases: active chat, no customer messages, and failed care.

Acceptance: accurate output, correct access and safe actions.

System Test Results

Test area	Case tested	Expected behavior	Observed result	Status
Access and privacy	Agent, Customer, Supervisor	Agent sees only assigned cases. Customer sees no internal data. Supervisor sees management tools.	Agent opened assigned case; unassigned case was not visible. Customer portal showed no scores, AI drafts or internal tickets. Supervisor had Dashboard and Assignment.	PASS
Explanation / message	No-customer-messages case (8443)	AI drafts for staff review. Nothing reaches the customer without staff approval.	Explanation shown to staff only. Message draft placed in the composer; not sent until staff clicked Send.	PASS
Call script	Active-chat, No-customer-messages, Failed-care (2905)	Scripts can be generated on open cases. New AI requests are blocked once a case is closed.	Scripts generated on open cases, including a staff-guided rewrite. New request blocked on failed case 2905.	PASS
Ticket and clarification	Active-chat case (8146, Supervisor)	AI drafts a ticket from chat evidence and lists missing details. Staff decide whether to ask the customer or submit the ticket.	Ticket drafted with quoted chat evidence; clarification message drafted. Neither was sent or submitted automatically.	PASS
Empty-chat protection	No-customer-messages case (8443)	AI ticket drafting is unavailable when there are no customer messages to cite.	“Draft ticket from chat” stayed disabled until customer messages existed; manual draft remained available.	PASS
Closed-case sending guard	Failed care (2905)	Send is disabled in the workspace once a case is closed (Failed or Care completed).	Send to simulated Zalo stayed disabled on the failed case, even with a non-empty draft.	PASS
Reply and care outcome	Active-chat (8146); Failed-care (2905)	Customer reply is saved and shown to staff. Failed care does not mean the customer left.	Reply persisted after reload and appeared in the Supervisor chat. Failed case kept retention outcome Unknown.	PASS
Supervisor dashboard	October cohort, one Agent	Filters and drill-down counts match. Outcomes stay unconfirmed until the period ends.	Agent had 51 October cases, 40 open; drill-down listed 40. All 51 outcomes showed “Awaiting period end”.	PASS
Assignment preview	Manual, Round-robin and Weighted	Supervisor previews and confirms assignments within capacity. Ownership, workload and Agent access update correctly.	Assignments respected capacity and persisted after reload. Workload, notifications and Agent access reflected the confirmed owner.	PASS

BUSINESS VALUE & STRATEGY

Expected Business Value

Illustrative nationwide, full-coverage scenario (Year 1 of operation) · Existing prediction model, SHAP and customer-care platform reused. Not the Bình Dương pilot.

BENEFIT — Additional customers retained

Assumption	Existing process	Proposed solution
Unique high-risk customers identified/month (national, assumed)	90,000	**100,000**
Customers receiving timely care/month	50,000	**70,000**
Continued-service rate among cared customers (assumed)	60%	**62%**
Continued-service rate without timely care	**30%** assumed · **62%** for newly cared is assumed; to be validated by A/B in Wave 1	

Customer group	Customers /month	Existing process	Proposed solution	Additional retained
Already receiving timely care	50,000	60% continue	62% continue	**1,000**
Newly receiving timely care	20,000	30% continue	62% continue	**6,400**
Total	**70,000**			**7,400**

Value mechanism: AI-assisted preparation frees agent capacity, extending timely care from 50,000 to 70,000 customers/month. Most of the benefit comes from this added coverage.

4.75 months is the average paid period recognised in Year 1 of operation, assuming customers are retained evenly throughout the year, with up to six additional paid months each.

Scope: Figures model a nationwide steady-state scenario (≈100,000 high-risk customers/month), not the Bình Dương pilot (≈2,500 churners/month, 60 agents).

Key assumptions: 62% continued-service for newly cared customers (+32 pts) drives ~86% of benefit; validate via A/B and permanent holdout (Waves 1–3). Care contact does not by itself prove retention.

Additional retained customers: 7,400/month · 88,800/year
Equivalent to an **11.38%** relative reduction in baseline churn (assumes ≈780,000 baseline churners per year nationally).
Monthly contribution: VND 250,000 (assumed national average ARPU) × 40% = VND 100,000 per additional retained customer.
Year-1 benefit: 88,800 customers × VND 100,000/month × 4.75 months = VND 42,180 million.

Expected Cost and ROI

Illustrative nationwide, full-coverage scenario (Year 1 of operation) · Existing prediction model, SHAP and customer-care platform reused. Not the Bình Dương pilot.

COST — Initial development and 12-month operation

Cost item	Calculation	Cost (Million)
Data/ML adaptation	40 hours × 500,000	**20**
Software integration	240 hours × 400,000	**96**
Testing and onboarding	80 hours × 250,000 + 80 hours × 150,000	**32**
Incremental incentives	(70,000 cus x 12) × 20% redemption × VND 150,000	**25,200**
AI and communication	(70,000 cus x 12) × 2,000	**1,680**
Additional scoring and infra	5 million/month × 12	**60**
Maintenance and monitoring	40 hours/month × 12 × 300,000	**144**
Total Year-1 cost	**Initial development: 148 + Annual operation: 27.084**	**27,232**

Cost basis: Incremental costs only. Existing CS agent labour is excluded (no added headcount); this assumes AI-assisted preparation (target −30% review time, Gate 1) absorbs +20,000 customers/month within current capacity, to be validated in pilot.

RETURN — Benefit after all incremental costs

Break-even is reached at 65% of the base-case uplift, a 35% margin of safety.
It also holds if newly cared customers reach ≈49% continued-service (vs. 62% assumed), other inputs unchanged.

Scenario	Additional retained customers / year	Year-1 contribution benefit (Million)	Net value (Million)	ROI
Break-even (65% of Base)	~57,331	~27,232	**~0**	**~0%**
Base (100%)	7,400 x 12 = 88,800	88,800 × 100,000 × 4.75 = 42,180	**14,948**	**54.9%**
High (150%)	133,200	63,270	**36,038**	**132.3%**

BUSINESS VALUE & STRATEGY

Pilot Evaluation and Success Metrics

To evaluate whether our solution creates real business value, we propose a randomized pilot comparing the AI-assisted workflow against the current retention process.

01 / PILOT DESIGN

Randomized comparison against the current retention process

ELIGIBLE COHORT

At-risk customers eligible for retention contact

RANDOM ASSIGNMENT

Customer-level split, fixed across scoring batches

GROUP A: NEW WORKFLOW — AI suggestions + staff review and approval

GROUP B: CURRENT PROCESS — Existing retention care procedure

DESIGN CONTROL — Stratify by risk and value · Keep assignments fixed across scoring cycles · Analyze by assigned group (ITT)

02 / SUCCESS METRICS

Prioritize causal business impact over system performance

PRIMARY BUSINESS OUTCOME

Incremental retention

Treatment retention – Control retention

Use mature churn labels; report percentage-point uplift and 95% confidence interval.

01 / NET BUSINESS VALUE

- Incremental net contribution per customer
- Contact, offer and AI costs

02 / OPERATIONAL EFFICIENCY

- Time to first contact
- Staff handling time per case

03 / AI ASSISTANCE

- Suggestion accept/edit rate
- Factual or inappropriate suggestions

04 / GUARDRAILS

- Opt-outs and complaints
- Inappropriate offers or contact

MODEL READINESS

GO / NO-GO DECISION — Scale only if predefined retention and net-value targets are met, with no guardrail breaches.

BUSINESS VALUE & STRATEGY

Implementation Roadmap

STAGE 1 · CORE PRODUCT ENGINEERING

Phased feature rollout and controlled nationwide scaling

01 Data & Features

- **Rolling-month windows**

Updated days 1, 11, 21

- **127 behavior metrics**

Usage, payment, tickets

- **Purged train splits**

Zero temporal leakage

02 Tune LightGBM

- **Validation AUC: 0.9483**

Test AUC: 0.9467

- **Platt calibrated**

Customer-grouped OOF

- **Risk threshold >500**

Alerts high-risk batch

03 SHAP Risk

- **Native TreeSHAP**

Signed contributions (log-odds)

- **6 approved groups**

Network, Billing, CS, Usage,...

- **Top-3 Drivers**

Normalized net impact

04 Care Queue & AI

- **Priority queue UI**

High risk >500 assignment

- **AI explanation agent**

Translates SHAP to plain text

- **Grounded facts**

Zero hallucination guardrail

05 Outreach Review

- **AI draft generation**

Personalized Zalo & call script

- **Offer catalog check**

Strict eligibility checks

- **Mandatory approval**

Staff edit, rewrite & sign-off

06 Chat-to-Ticket

- **AI ticket extraction**

Parses chat into tech ticket

- **Smart reminders**

3h/24h follow-up alerts

- **Cloud database**

28 tables, 32 RPCs, RLS

07 Prototype QA

- **End-to-End Testing**

73 DB & UI checks passed

- **n8n quota guard**

Controlled exec budget <1000

- **Security & Access**

Agent vs Supervisor RLS

PRE-PILOT GATE: Enterprise Data & Policy Sign-off

STAGE 2 · PHASED PILOT IN BINH DUONG

Sequential feature-by-feature rollout with Gate Review & Review Cycles

P1 Feature 1: Diagnostic AI

- Roll out Care Queue & SHAP
- Read-only AI explanation before regular calls
- Staff validates customer risk drivers

TARGET METRICS & GATE 1:

- Pre-call case review time cut by >30%
- AI diagnostic factual accuracy >90%

P2 Feature 2: Assisted Outreach

- AI generates personalized Zalo/call draft
- Mandatory staff review, rewwrite and approval
- Controlled A/B pilot vs manual standard care

TARGET METRICS & GATE 2:

- Staff draft acceptance/edit rate >70%
- Delivery & reply rate rises; opt-out rate <2%

P3 Feature 3: Chat-to-Ticket

- AI parses network complaints to Tech Ticket
- Automated 3h/24h follow-up SLA reminders
- Tracks true 30-day period-end retention

TARGET METRICS & GATE 3:

- 0 tickets returned for missing evidence
- Fast handoff; resolved cases churn decreases

REVIEW

NO-GO

Revise/Stop

GO

STAGE 3 · ENTERPRISE EXPANSION & SCALING

Controlled multi-province expansion, continuous training, and break-even governance

W1 Wave 1: 3 to 5 Provinces

- Test beyond Binh Duong (mix high & moderate churn)
- Maintain permanent holdout control group for 2-3 months
- Verify if SHAP drivers & message acceptance replicate

W2 Wave 2: 10 to 15 Provinces

- Stress-test system concurrency & live scoring feeds
- Audit total intervention cost vs retained customer margin
- Monitor LLM token budgets & n8n execution efficiency

W3 Wave 3: Nationwide Rollout

- Deployed to all nationwide customer care centers
- Direct integration into enterprise core CRM & ticketing
- Keep 5% permanent holdout control for ongoing ROI audit

C Continuous Learning

- Monthly outcome feedback

Active vs churned ground truth

- Retrain LightGBM

Dynamic drift adaptation

Key Risks

ID	Risk	Description	L x I	Decision
R1	**Privacy, consent and channel rules breached**	Contacting opted-out customers, breaking Zalo rules, duplicate contact (predictions refresh on days 1, 11, 21), or chat data sent to an outside AI provider.	**4 x 5 = 20 Critical**	**Avoid**: recheck consent at approval and delivery; one owner per case; privacy sign-off before launch.
R2	**Unsupported AI content**	The AI may invent debt, discounts or reasons, or treat SHAP as proof. The prototype writes English only; its knowledge base is demo FAQ.	**2 x 3 = 6 Medium**	**Mitigate**: staff approve every message; approved offers and FAQ only; native-speaker review; pause if errors rise.
R3	**Unproven economics**	Break-even needs 65% of the base-case uplift (≈6.8 pts of extra retention on 70,000 cared customers). The uplift is assumed, and high-risk customers are not always persuadable	**4 x 4 = 16 High**	**Accept (capped)**: fixed pilot budget; validate uplift by A/B test and holdout; no-go if uplift is below break-even.
R4	**Staff capacity and adoption**	Staff review every AI draft, and may skip a tool that is hard to use.	**3 x 3 = 9 Medium**	**Mitigate**: cap the list to capacity; rank overdue cases first; training and usability test.
R5	**Policy and ownership decisions pending**	Offer catalog, follow-up delay, CS sponsor and owners are not set. A technical handoff proves a request was sent, not that the fault was fixed.	**4 x 3 = 12 High**	**Mitigate**: Stage 2 gate with named owners; Finance-approved offer catalog; track technical handoffs.
R6	**Access-control or security failure**	22 permission-bypassing functions are flagged, leaked-password protection is off and concurrent use is unproven. Not a production sign-off.	**2 x 4 = 8 Medium**	**Mitigate**: review flagged functions; enable password protection; run access and double-approval tests.

Risk priority matrix (5 x 5)



Likelihood: 1 Rare, 2 Unlikely, 3 Possible, 4 Likely, 5 Almost certain

Impact: 1 Negligible, 2 Minor, 3 Moderate, 4 Major, 5 Severe

Low (1-4) | Medium (5-9) | High (10-16) | Critical (20-25)

THANK YOU

Group 3

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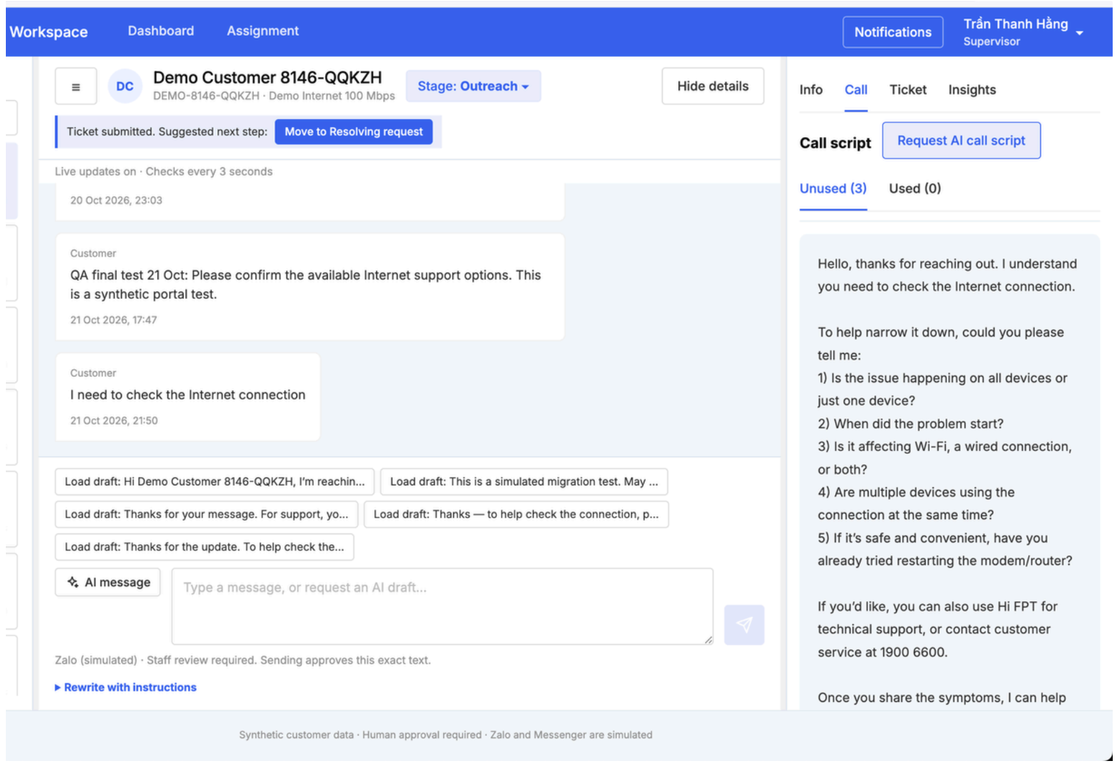
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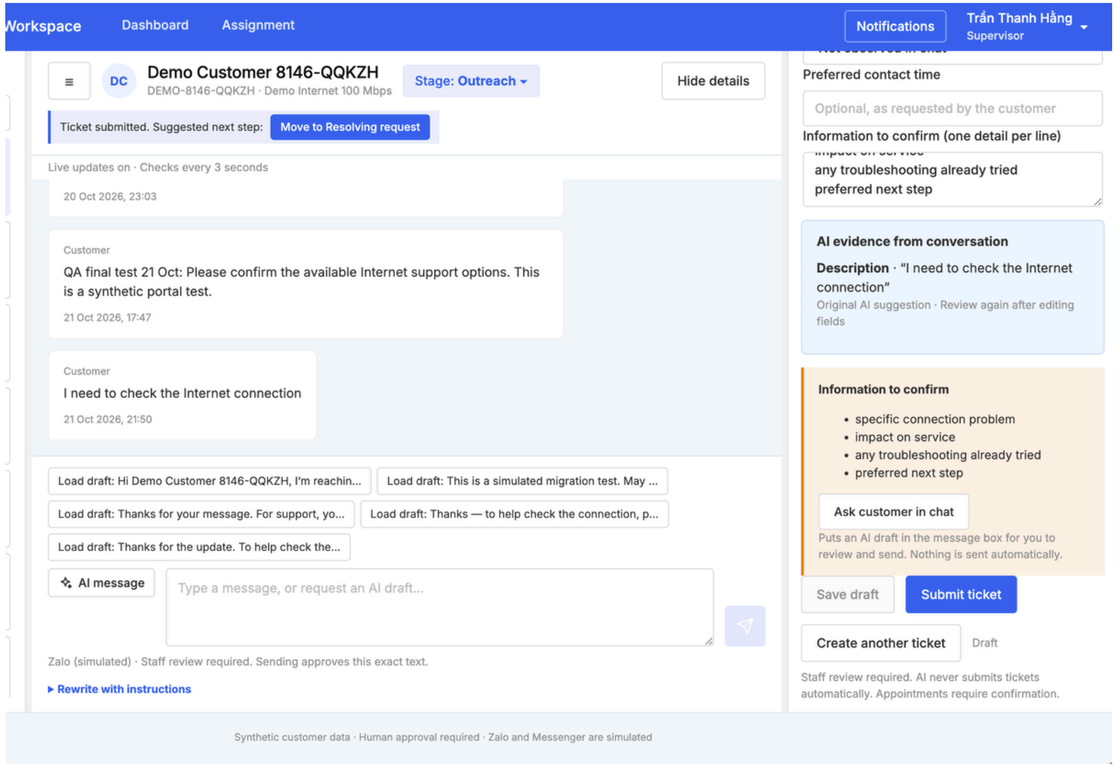
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Appendix A: Case: Active Chat

A1 Call script follows the latest request



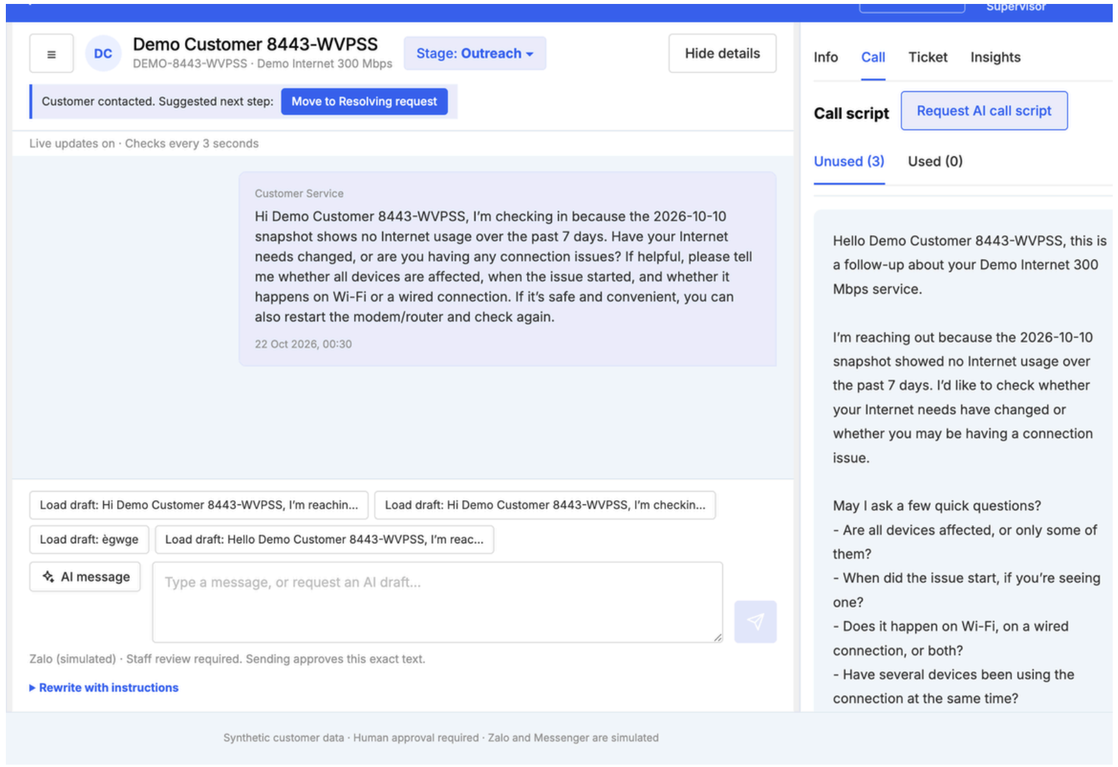
A2 Ticket evidence and missing details



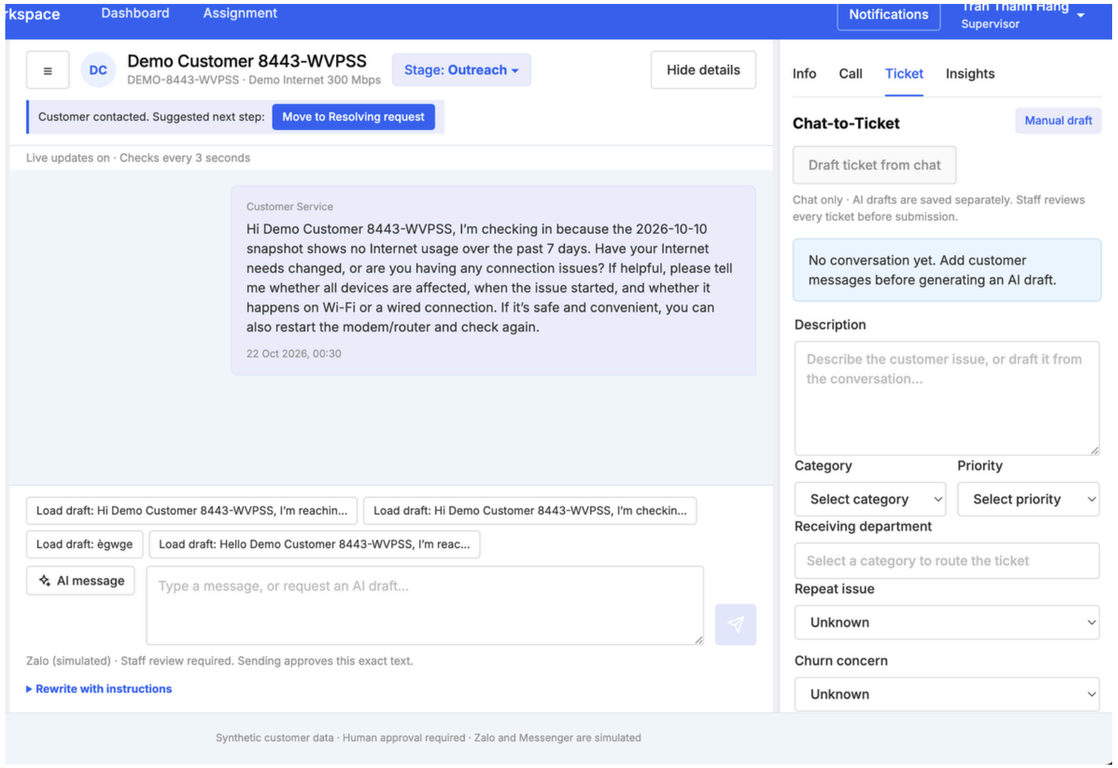
Technical Support draft. No submission.

Appendix B: Case: No customer messages

B1 Call script after staff-guided rewrite



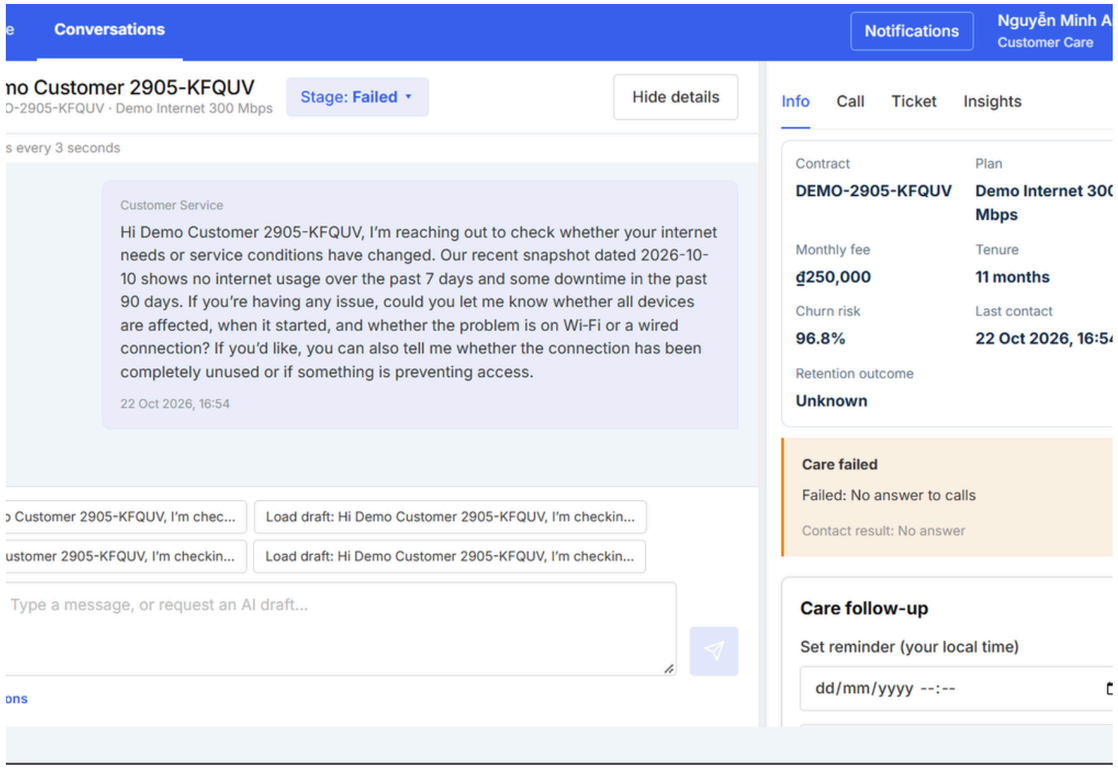
B2 No customer messages, AI ticket drafting blocked



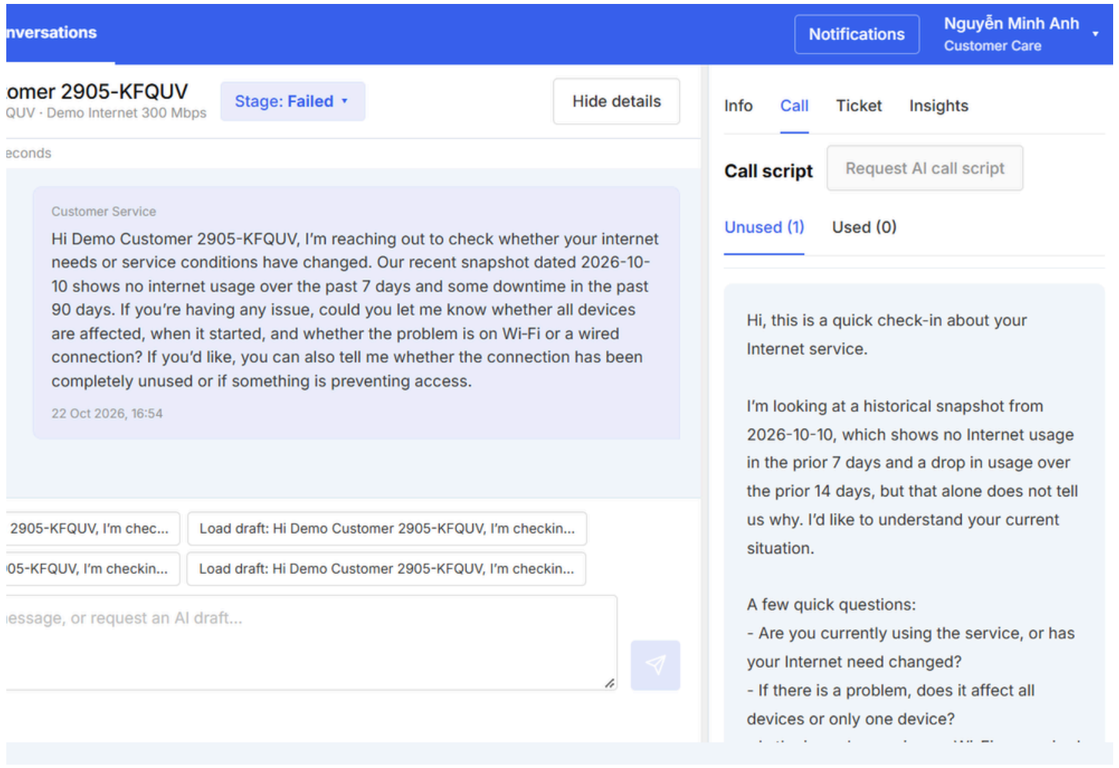
Missing renewal timing stays unverified

Appendix C: Case: Failed-Care

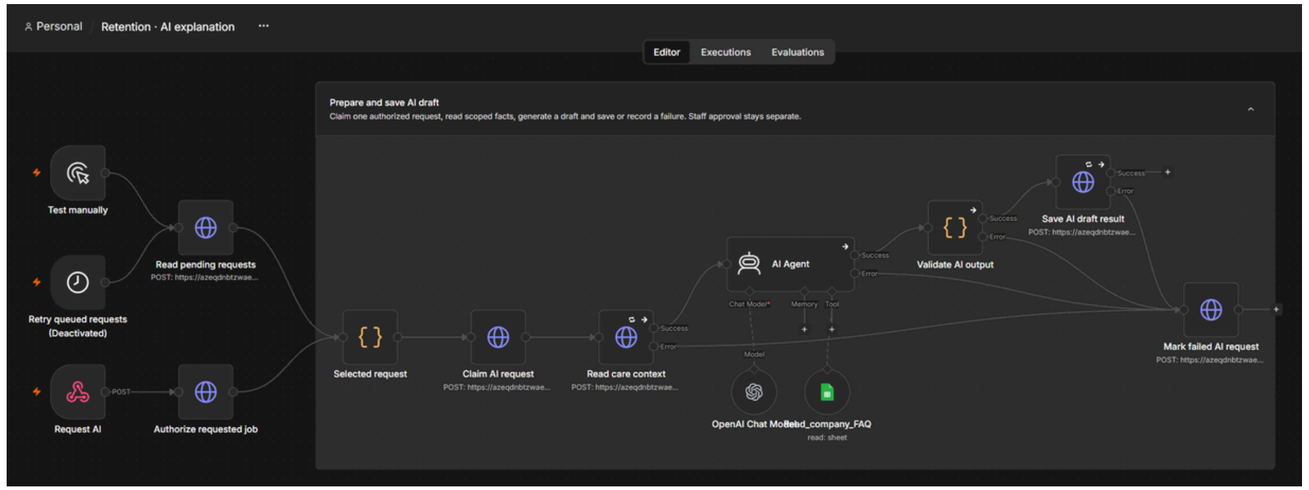
C1 Care failed (no answer to calls & messages)

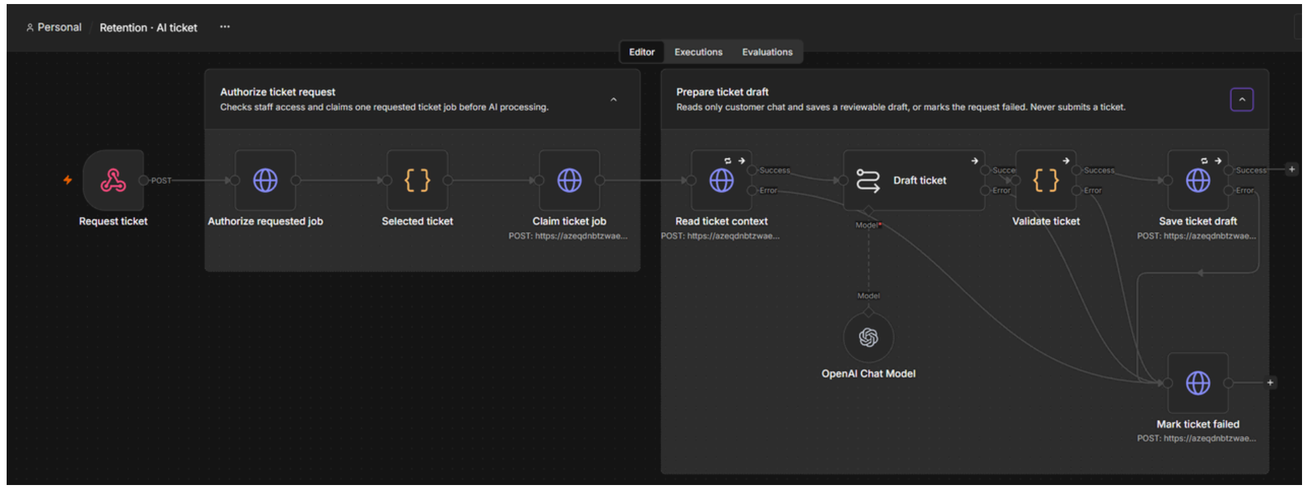


C2 New Call script request disabled



Appendix D: n8n AI Agent Workflows





Appendix D: n8n AI Agent Workflows



